



“ THE **SHAPE** OF THE **EARTH** IS NOT **FLAT**, AS SOME SUPPOSE WHO IMAGINE IT TO BE LIKE AN **EXPANDED DISK**...”

Martianus Capella, from *On the Marriage of Philology and Mercury*, 410–439

7 THE NUMBER OF LIBERAL ARTS IDENTIFIED BY ROMAN WRITER CAPELLA

of right-angled triangles and in around 300, **Sun Zi** compiled the *Sun Zi Suan Ching*, which includes an analysis of indeterminate equations. It also contains what is now known as the **Chinese remainder theorem**, which provides a method of finding solutions to problems in modular arithmetic (also called clock arithmetic, because numbers are arranged in a circle, rather than along the number line). In the 5th century, **Zu Chongzhi** (429–500) wrote *Zhui Shu* (*Method of Interpolation*), in which he **calculated pi** to be $\frac{355}{113}$. He refined this to produce a value for pi that was accurate to seven

decimal places (see panel, below), a figure that was not improved upon until the 16th century.

Martianus Capella, from Madaura in North Africa, established the basic structure of early medieval European education. In his *On the Marriage of Philology and Mercury* (410–439), he presented a **compendium of knowledge**, which he divided into the trivium (grammar, dialectic, and rhetoric) and the quadrivium (geometry, arithmetic, astronomy, and music). In this work, he stated that **Mercury and Venus orbit around the Sun**, a view that Copernicus used to support his

heliocentric view of the solar system (see 1543).

Mathematics progressed only slowly during the later Roman empire. In about 450, the Neoplatonist philosopher **Proclus** (c.410–485) produced his *Commentary on Euclid*, in which he preserved the work of earlier mathematicians. Proclus's contemporary, **Domininus of Larissa** (c.410–480) wrote *Manual of Introductory Arithmetic*, which included a summary of number theory.

By the 5th century, the Maya had devised a **sophisticated calendrical system** and a notation system for numbers that could express any number using only three symbols: a dot for 1, a bar for 5, and a shell for 0. Maya astronomers were particularly concerned with lunar cycles, the Sun, eclipses, and movements of the planet Venus.

There is also evidence that the Maya were practicing **raised-field agriculture** from as early as the mid-3rd century to utilize fertile land that would otherwise have been too waterlogged for agricultural use.

Astronomical codex
A section from the Dresden Codex, a 9th-century Maya astronomical work that includes detailed tables for movements of the planet Venus.



THE VALUE OF PI

Pi, the ratio of the circumference of a circle to its diameter, was estimated at 3.125 by the Babylonians. The Greeks discovered a method of calculating it by using the sides of a polygon inside a circle to approximate the circumference, and Archimedes used this method to give a figure of $\frac{22}{7}$. In about 475, Zu Chongzhi calculated pi as 3.1415926—accurate to seven decimal places. Computers have now calculated this value to trillions of decimal places.



c.360 Oribasius of Pergamum writes on **dietetics**

c.400 Theon of Alexandria introduces a method of calculating **square roots** by approximation

410–439 Martianus Capella asserts that **Mercury and Venus orbit around the Sun**

c.450 Greek philosopher Proclus writes his *Commentary on Euclid*

mid- to late 400s Domininus of Larissa writes his *Manual of Introductory Arithmetic*

c.475 Zu Chongzhi calculates pi to seven decimal places

c.499 Indian mathematician Aryabhata (476–550) estimates pi to be 3.1416